

Electro Chemistry

Farade's Law

First Law

The amount of a given product liberated at an electrode during electrolysis is directly proportional to the quantity of electricity which passes through the electrolyte solution.

Second Law

When the same quantity of electricity passes through solutions of different electrolytes, the amounts of the substances liberated at the electrodes are directly proportional to their chemical equivalents.

Conductance

Specific conductance

It is evident that a substance which offers very little resistance to the flow of current allows more current to pass through it. Thus the power of a substance to conduct electricity or conductivity is the converse of resistance. The reciprocal of specific resistance is termed Specific conductance or Specific conductivity.

Equivalent Conductance

It is defined as the conductance of an electrolyte obtained by dissolving one gram-equivalent of it in V cc of water.

Diff between Ionization vs dissociation

Ionization	Dissociation
1. It is the process Which produces new Changed particles. 2. It involves polare Covalent compounds or metals. 3. Irreversible. 4. Involve Covalent bond between atoms. 5. Always Produces Charged particles.	1. It is the separation of charged Particles which already exists in a compound. 2. It involves ionic Compounds. 3. Reversible 4. Involve ionic bond in compound. 5. St produces either Charged Particles or electrically neutral Particles.

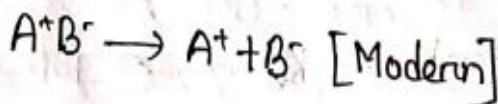
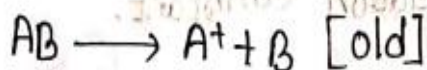
Arrhenius Theory of Ionization:-

Savant Arrhenius theory studied the conduction of current through water solution of electrolytes. He came to believe that the conductivity of solutions was due to the presence of ions.

Arrhenius theory of ionization may be stated as:-

① When dissolved in water, neutral electrolyte molecules are split up into two types of charged particles. These particles are called ions and the process was termed as ionization. The positively charged are called cation and the negatively charged are called anion.

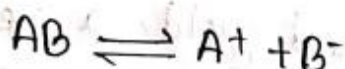
In this modern form, the theory assumes that the ions are already present in the solid electrolytes and ~~are~~ these are held together by electrostatic force. When placed in water, these neutral molecules dissociate to form separate cations and anions.



For these reason, this theory may be referred to as theory of electrolytic dissociation.

② The ions present in solution constantly reunite to form neutral molecules.

Thus there is a state of equilibrium between the dissociated molecules and ions.



Applying the law of mass Action to the ionic equilibrium, we have

$$K = \frac{[A^+][B^-]}{[AB]}$$

where, K is called the dissociation constant.

WHAT IS TRANSPORT NUMBER ?

During electrolysis the current is carried by the anions and the cations. The fraction of the total current carried by the cation or the anion is termed its Transport number or Hittorf's number. If v_+ represents the speed of migration of the cation and v_- that of the anion,

$$\text{the transport number of cation} = \frac{v_+}{v_+ + v_-}$$

$$\text{the transport number of anion} = \frac{v_-}{v_+ + v_-}$$

The transport number of the cation is represented by t_+ and that of the anion by t_- .

$$\text{Thus } t_+ = \frac{v_+}{v_+ + v_-} \quad \text{and} \quad t_- = \frac{v_-}{v_+ + v_-}$$

$$\text{or } \frac{t_+}{t_-} = \frac{v_+}{v_-} \quad \text{and} \quad t_+ + t_- = 1$$

If the speed ratio v_+/v_- be denoted by r , we have,

$$r = \frac{t_+}{t_-} = \frac{t_+}{1 - t_+}$$

$$\text{and } t_- = \frac{1}{1 + r}$$

SOLVED PROBLEM 2. The resistance of decinormal solution of a salt occupying a volume between two platinum electrodes 1.80 cm apart and 5.4 cm² in area was found to be 32 ohms. Calculate the equivalent conductance of the solution.

SOLUTION

Here $l = 1.80$ cm and $A = 5.4$

$$\therefore \text{cell constant } x = \frac{l}{A} = \frac{1.80}{5.4} = \frac{1}{3}$$

$$\text{observed conductance} = \frac{1}{32} \text{ mhos}$$

Since the solution is $N/10$, $V = 10,000$ ml

Now, specific conductance = $x \times \text{obs. conductance}$

874 24 PHYSICAL CHEMISTRY

or

$$\kappa = \frac{1}{3} \times \frac{1}{32} = \frac{1}{96} \text{ mhos}$$

$$\Lambda = \kappa \times V$$

$$= \frac{1}{96} \times 10,000$$

$$= 104.1 \text{ mhos cm}^2 \text{ equiv}^{-1}$$

SOLVED PROBLEM 1. A solution of silver nitrate containing 12.14 g of silver in 50 ml of solution was electrolysed between platinum electrodes. After electrolysis, 50 ml of the anode solution was found to contain 11.55 g of silver, while 1.25 g of metallic silver was deposited on the cathode. Calculate the transport number of Ag^+ and NO_3^- ions.

SOLUTION

Weight of Ag in 50 ml of the solution before electrolysis = 12.14 g

Weight of Ag in 50 ml of the solution after electrolysis = 11.55 g

$$\begin{aligned}\therefore \text{Fall in concentration of Ag} &= 12.14 - 11.55 \\ &= 0.59 \text{ g} \\ &= 0.0055 \text{ g eq}\end{aligned}$$

Weight of Ag deposited in silver coulometer = 1.25 g

$$\begin{aligned}&= \frac{1.25}{108} \text{ g eq} \\ &= 0.0116\end{aligned}$$

$$\begin{aligned}\text{Hence, transport number of } \text{Ag}^+ (t_{\text{Ag}^+}) &= \frac{\text{Fall in conc. around anode}}{\text{No. of g eqvt deposited in silver coulometer}} \\ &= \frac{0.0055}{0.0116} \\ &= 0.474\end{aligned}$$

$$\begin{aligned}\text{and Transport number of } \text{NO}_3^- \text{ ion } (t_{\text{NO}_3^-}) &= 1 - 0.474 \\ &= 0.526\end{aligned}$$

SOLVED PROBLEM 2. In an electrolysis of copper sulphate between copper electrodes the total mass of copper deposited at the cathode was 0.153 g and the masses of copper per unit volume of the anode liquid before and after electrolysis were 0.79 and 0.91 g respectively. Calculate the transport number of the Cu^{2+} and SO_4^{2-} ions.

SOLUTION

Wt. of copper in the anode liquid before electrolysis = 0.79 g

Wt. of copper in the anode liquid after electrolysis = 0.91 g

$$\begin{aligned}\text{Increase in weight} &= 0.91 - 0.79 \\ &= 0.12 \text{ g}\end{aligned}$$

Increase in weight of copper cathode in the coulometer = 0.153 g

This means that if no copper had migrated from the anode, increase in weight would have been 0.153 g.

$$\text{But actual increase} = 0.12$$

$$\begin{aligned}\text{Fall in concentration due to migration of } \text{Cu}^{2+} \text{ ions} &= 0.153 - 0.12 \\ &= 0.033\end{aligned}$$

$$\therefore \text{Transport number of } \text{Cu}^{2+} \text{ ion} = \frac{0.033}{0.153} = 0.215$$

$$\begin{aligned}\text{and Transport number of } \text{SO}_4^{2-} \text{ ion} &= (1 - 0.215) \\ &= 0.785\end{aligned}$$

☐ Hittorf's gave a general rule is known as Hittorf's Rule.

It states that:

"The loss of concentration around any electrode is proportional to the speed of ions moving away from it."

Illustration of Hittorf's Rule:

In the figure,

A is the anode, C is the cathode.

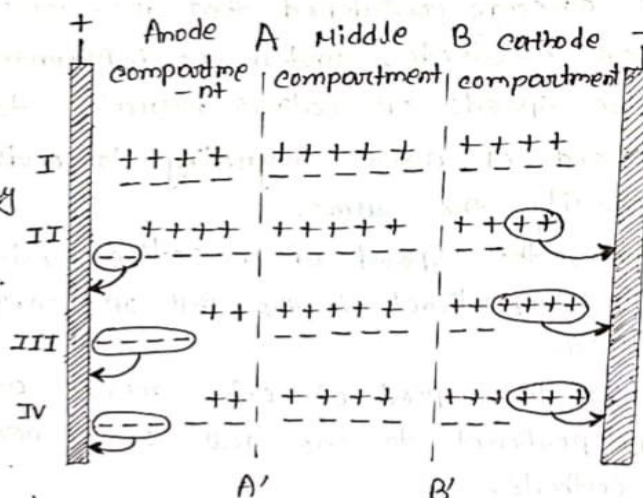
AA' and BB' are two imaginary planes which divide the cell into three compartments.

→ Anode compartment

→ Middle compartment

→ Cathode compartment

→ The sign (+) represents a cation and (-) represents an anion.



Before electrolysis, let there be 13 ion-pairs in the cell.

The number of ion-pairs in the two outer compartments is 4 each and there are 5 ion-pairs in the middle compartment (position I).

(i) Let the anions alone be capable of movement

When 2 anions have moved towards the anode, we get the position (II). The cations have not moved at all.

The number of discharged anions and cations is the same viz 2. The concentration in the anode compartment has not altered while in the cathode compartment it has fallen by two ion-pairs.

(ii) Let the anions and cations move at the same rate

When 2 ions of each type have crossed towards the opposite electrodes, we get the condition shown in (III). The number of discharged anions and cations is the same viz. 4.

Anode and cathode compartment has fallen to the same extent by 2 ion pairs.

(iii) Let the cations move at twice the speed of the anions

When cations have moved to the cathode compartment, one anion passes into the anode compartment. This state is shown in (IV). Total number of discharged anions and cations is again same. In the concentration, Cathode compartment has fallen by one ion-pair, Anode compartment has fallen by two ion pairs.